Hadronic Shower Profile from DOM Pulses: Problem and Two Solutions

1 Inference problem

Unknowns. A hadronic cascade with vertex $\boldsymbol{v}$, time t_v, direction $\hat{\boldsymbol{u}}$, energy E, and longitudinal profile along the axis s:

$$P(s) \approx A\, g(s; a, b), \qquad g(s; a, b) = \frac{s^{a-1} e^{-s/b}}{\Gamma(a)\, b^a}. \tag{1}$$

$P(s)$ is either the number of particles crossing depth s, $N(s)$, or the charged-track length per unit depth, $\ell(s)$. Cherenkov light is proportional to $\ell(s)$, so $\ell(s)$ is the profile the pulses measure. The target is

$$\boldsymbol{y} = (a,\, b), \qquad \text{or equivalently mean } m = ab,\ \text{width } \sigma = \sqrt{a}\, b. \tag{2}$$

Data. Photoelectron counts n_{dk} at DOM $d = 1, \ldots, N$ in time bin $k = 1, \ldots, K$.

Model. Each count is Poisson with expected value $\mu_{dk}(\boldsymbol{\theta})$, where $\boldsymbol{\theta} = (\boldsymbol{v}, t_v, \hat{\boldsymbol{u}}, E, a, b)$. DOMs and bins are independent given $\boldsymbol{\theta}$:

$$\ln \mathcal{L}(\boldsymbol{\theta}) = \sum_{d=1}^{N} \sum_{k=1}^{K} \Big[\, n_{dk} \ln \mu_{dk}(\boldsymbol{\theta}) - \mu_{dk}(\boldsymbol{\theta}) \,\Big]. \tag{3}$$

Goal.

$$p(a, b \mid \mathcal{D}) \;\propto\; \mathcal{L}(\boldsymbol{\theta})\, p(\boldsymbol{\theta}) \quad \text{(marginalized over geometry and } E\text{)}. \tag{4}$$

Labels from simulation. For each Geant4 hadronic shower e, fit g to its true profile:

$$(a_e, b_e) = \arg\min_{a,b}\ \mathrm{fit}\big(P_e(s),\, g(s; a, b)\big). \tag{5}$$

2 Solution A: learn the answer directly (Neptune)

A point transformer f_ϕ maps the hits straight to the profile parameters and their uncertainty:

$$\big(\hat{\boldsymbol{y}},\ \hat{\boldsymbol{s}}^2\big) = f_\phi(\mathcal{D}), \qquad \boldsymbol{y} = (\ln a,\, \ln b). \tag{6}$$

Training on M simulated hadronic showers minimizes the Gaussian negative log-likelihood:

$$\mathcal{J}(\phi) = \frac{1}{M} \sum_{e=1}^{M} \sum_{i=1}^{2} \left[\frac{(y_{e,i} - \hat{y}_{e,i})^2}{2\, \hat{s}_{e,i}^2} + \ln \hat{s}_{e,i} \right]. \tag{7}$$

Inference is a single forward pass. The labels must come from full hadronic-shower simulation, so that (a, b) varies from shower to shower at fixed E.

3 Solution B: fit a likelihood (CANDELA, retrained)

The current CANDELA kernel is trained on EM point cascades with PPC's mean profile. It must be retrained on hadronic cascades, with the profile as an extra input:

$$\text{kernel:}\quad (\boldsymbol{x}_d,\ E,\ a,\ b) \longmapsto \Big(\Lambda(\boldsymbol{x}_d; a, b),\quad \pi(t \mid \boldsymbol{x}_d; a, b)\Big), \tag{8}$$

where $\boldsymbol{x}_d$ is the geometry of DOM d relative to the shower, Λ is the expected light per GeV, and π is the arrival-time distribution. The training showers must sample (a, b) over its natural range, not only its mean at each energy.

The expected counts are then

$$\mu_{dk}(\boldsymbol{\theta}) = E\ \Lambda(\boldsymbol{x}_d; a, b) \int_{\text{bin } k} \pi(t - t_v \mid \boldsymbol{x}_d; a, b)\, dt\ +\ \nu_{dk}, \tag{9}$$

with ν_{dk} the noise. The fit is

$$(\hat{a}, \hat{b}, \hat{E}, \hat{\boldsymbol{v}}, \hat{\boldsymbol{u}}) = \arg\min_{\boldsymbol{\theta}}\ \big[-\ln \mathcal{L}(\boldsymbol{\theta})\big], \tag{10}$$

and the uncertainty comes from the curvature of $-\ln \mathcal{L}$ at the minimum:

$$\mathrm{Cov}(\hat{a}, \hat{b}) \approx \Big[\nabla^2_{(a,b)}\big(-\ln \mathcal{L}\big)\Big]^{-1}. \tag{11}$$

4 Comparison

	A: Neptune	B: CANDELA (retrained)
Learns	$\mathcal{D} \mapsto (a, b)$	light from a shower with profile (a, b)
Training data	hadronic showers with (a, b) labels	hadronic showers with varied (a, b)
Inference	one forward pass	minimize $-\ln \mathcal{L}$ per event
Uncertainty	predicted $\hat{\boldsymbol{s}}$ (check coverage)	curvature of $-\ln \mathcal{L}$